Big Data System Design (MBAI 5110G)
Winter-2026

Week 4: Big Data Processing & Management in Cloud

Lectured by:

Dilli P. Sharma, PhD

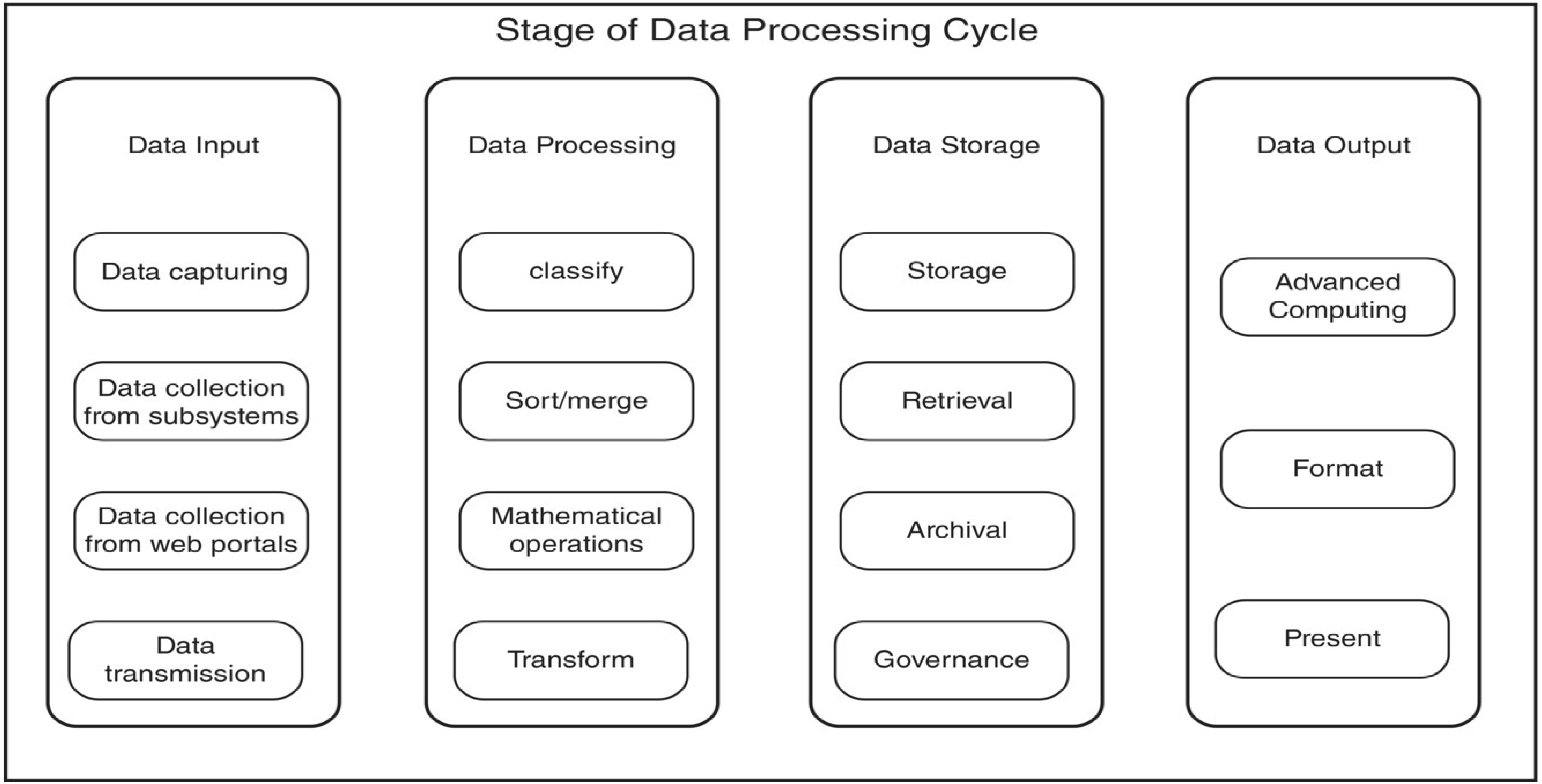
Outline

- Data Processing Concept
- Big Data Processing System Architectures
- Shared Everything Architecture
- Shared-Nothing Architecture
- Batch Processing
- Parallel Processing
- Distributed Computing
- Big Data and Server Virtualization
- Cloud Computing Concept
- Cloud Services and Architecture
- Google File System
- Cloud Storages
- Cloud Challenges

Data Processing

- Data processing is the process of **collecting, processing, manipulating, and managing data** to generate **meaningful information** for the **end user**.
- **Key Purpose:**
  - Transform raw data into useful insights
  - Support decision-making
  - Improve accuracy and efficiency

Data Processing Stages

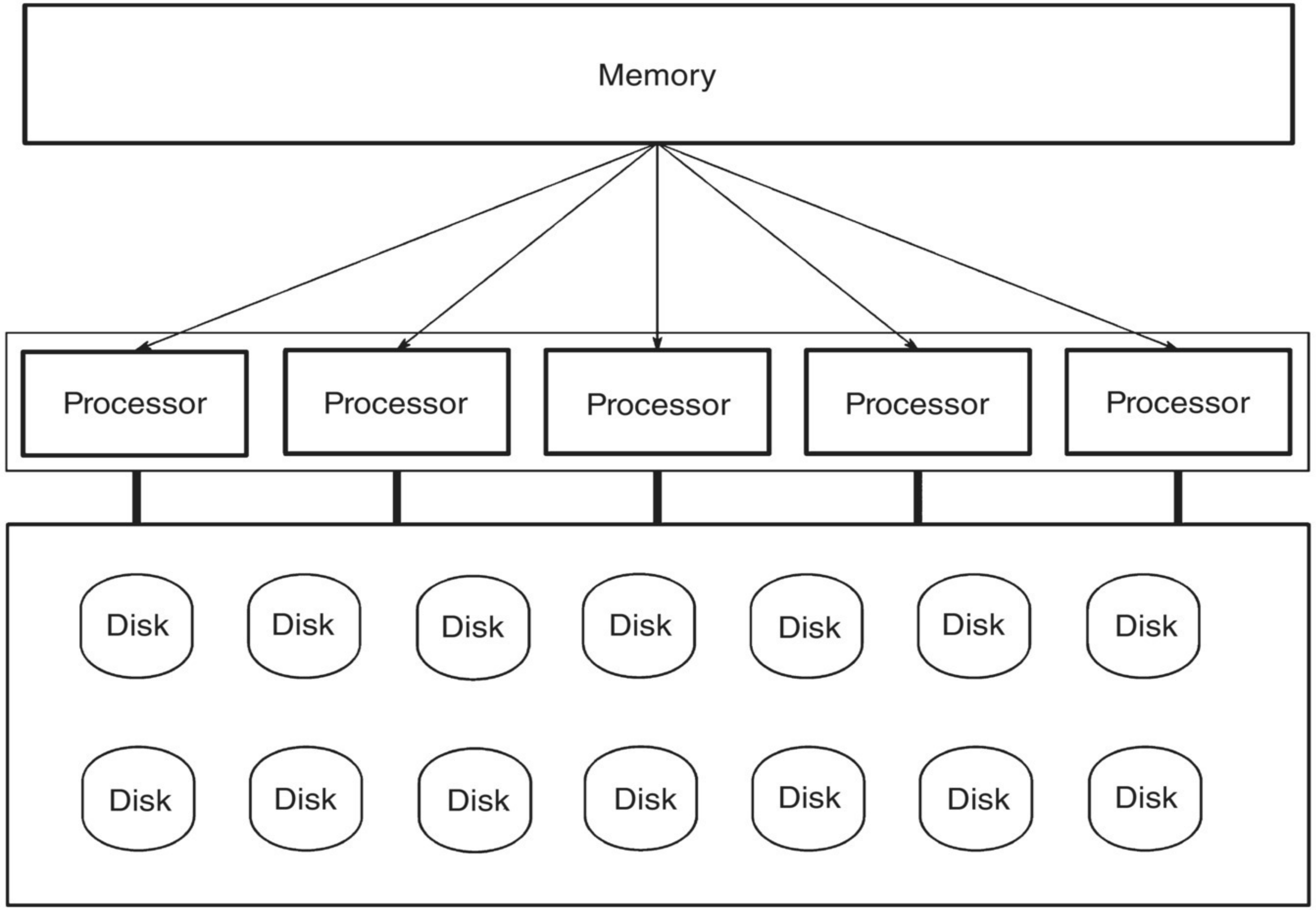


Big Data Processing System Architectures

- Defines how data and computation are organized
- Designed for scalability, fault tolerance, and performance

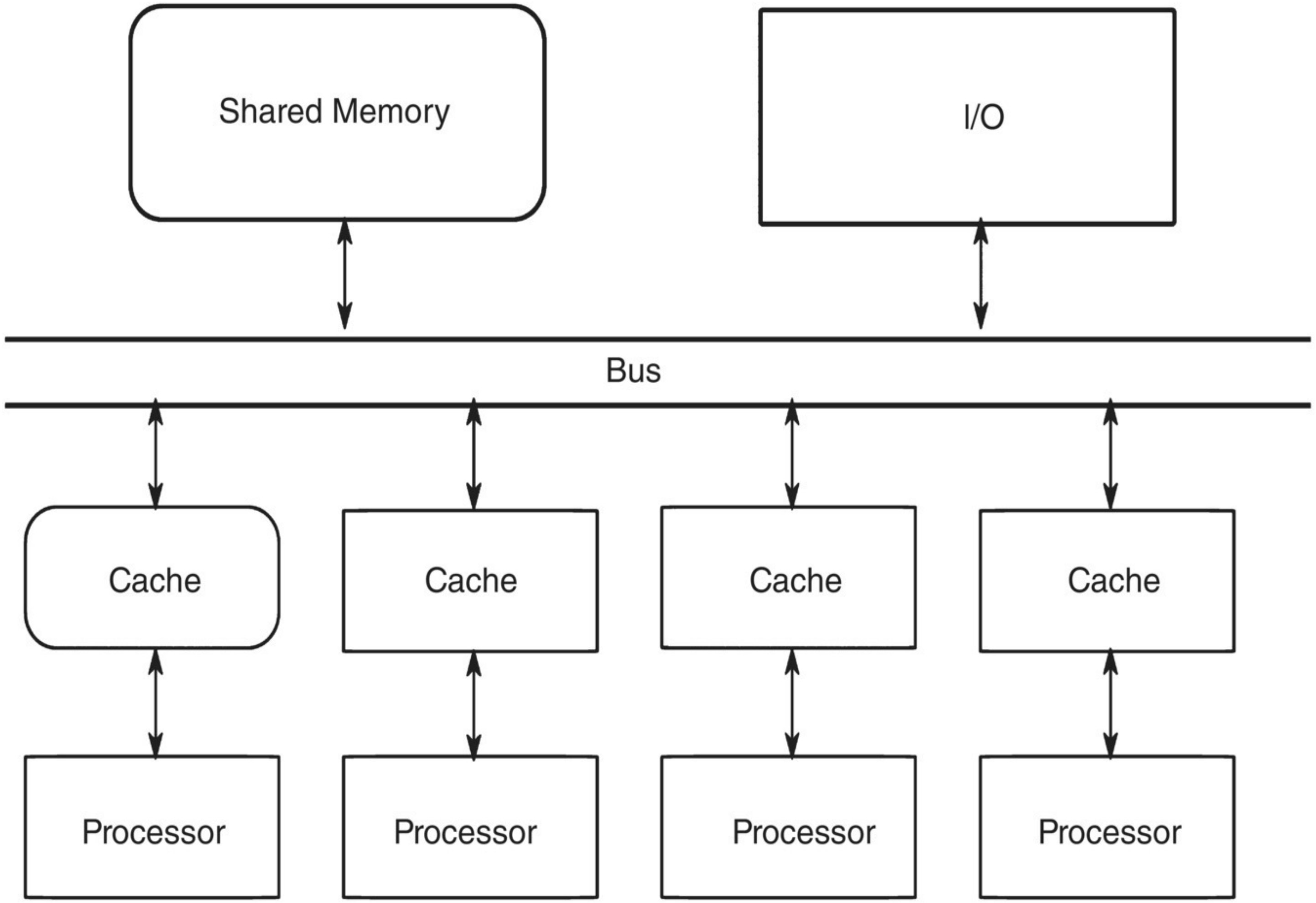
Shared-Everything Architecture

- All processors share memory and storage
- Simple to manage and program
- Limited scalability due to contention
- Two types:
  - Symmetric Multiprocessing (SMP)
  - Distributed Shared Memory (DSM)



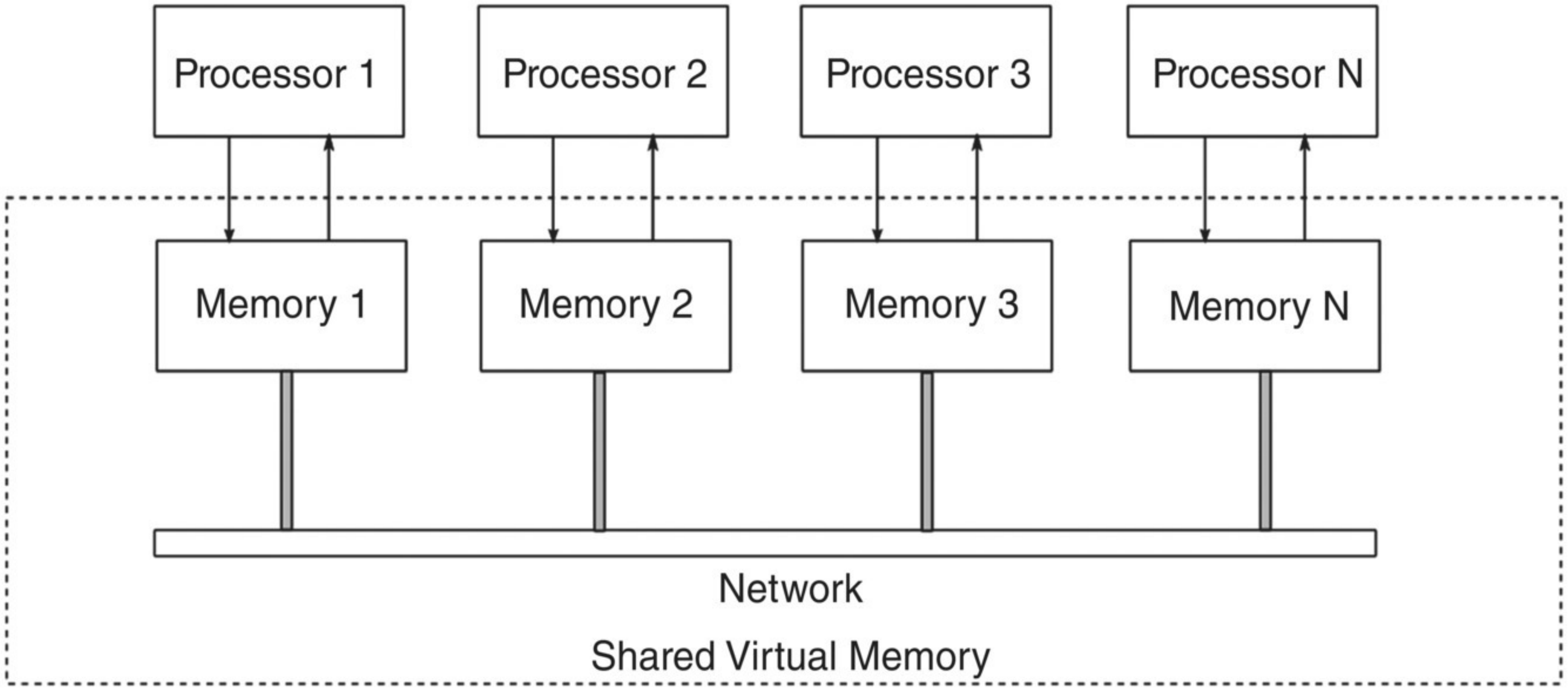
Symmetric Multiprocessing (SMP)

- Multiple processors share a single memory space
- Uniform memory access time
- Common in traditional servers



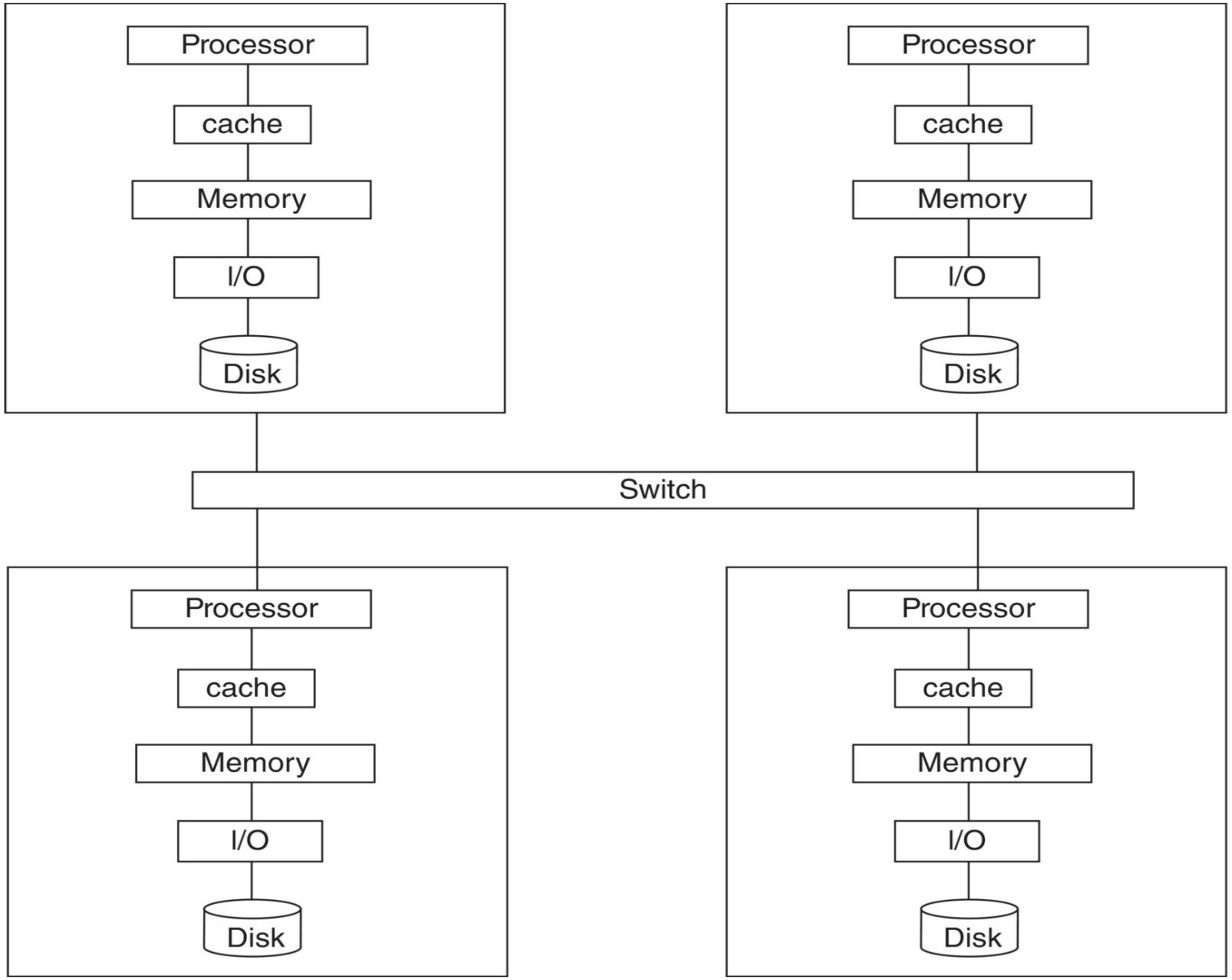
Distributed Shared Memory (DSM)

- Each node has its owr memory and storage
- Highly scalable and fault-tolerant
- Used in Hadoop and Spark systems



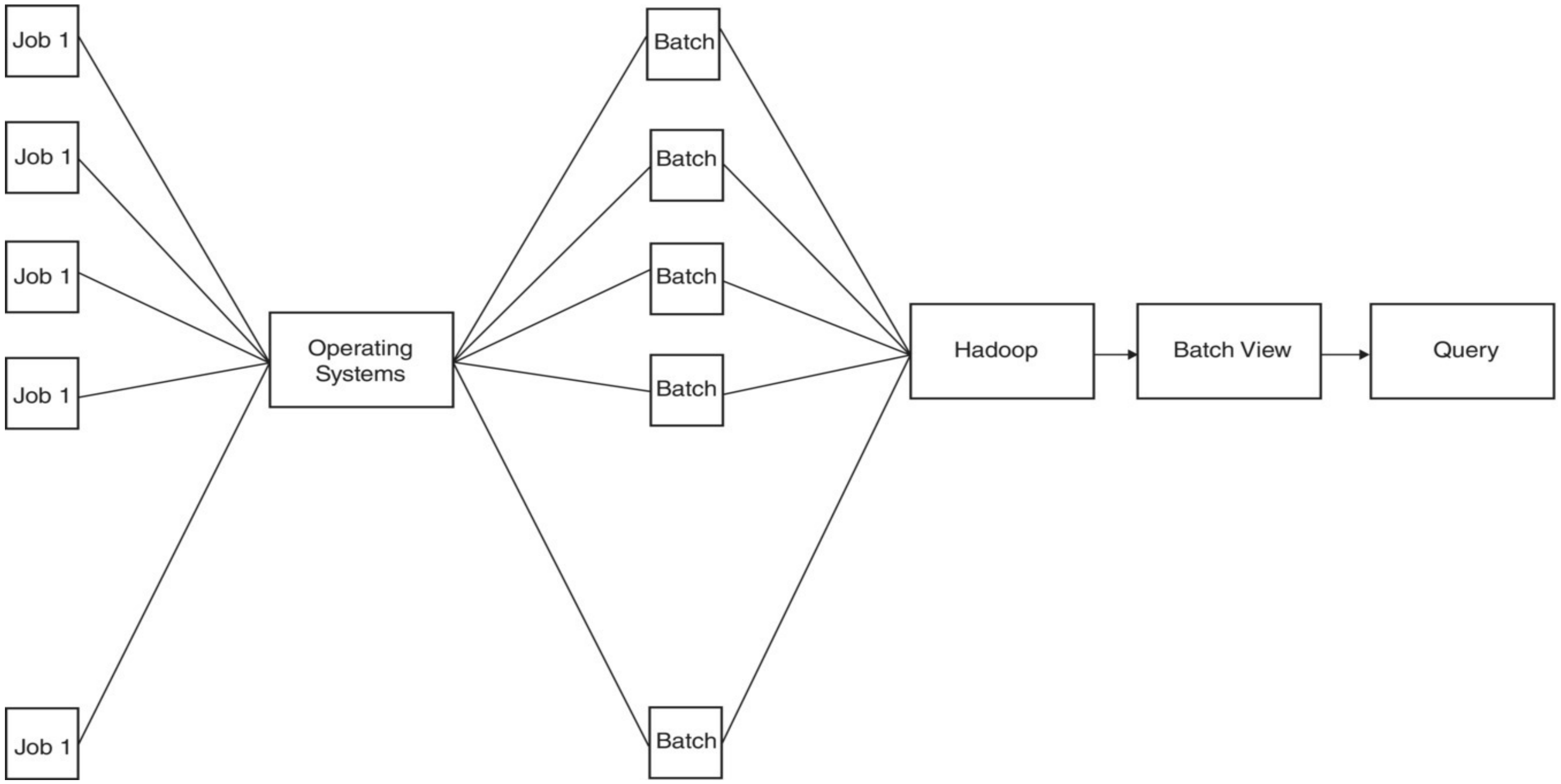
Shared-Nothing Architecture

- Each node has its own memory and storage
- Highly scalable and fault-tolerant
- Used in Hadoop and Spark systems



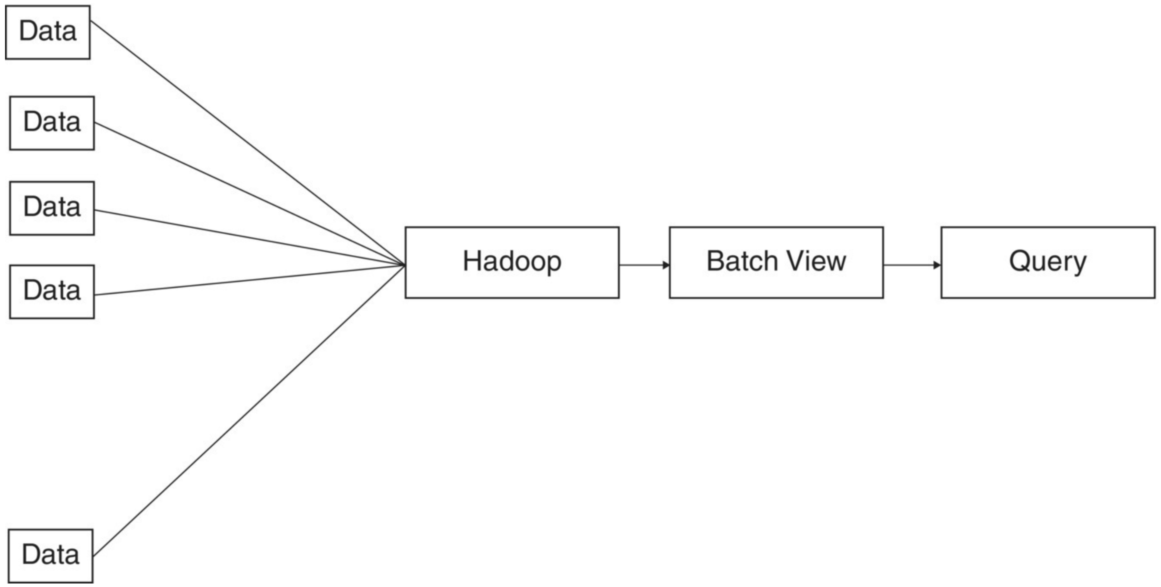
Batch Processing

- Processes large volumes of data at once
- High throughput, higher latency
- Example: Hadoop MapReduce



Real-Time Processing

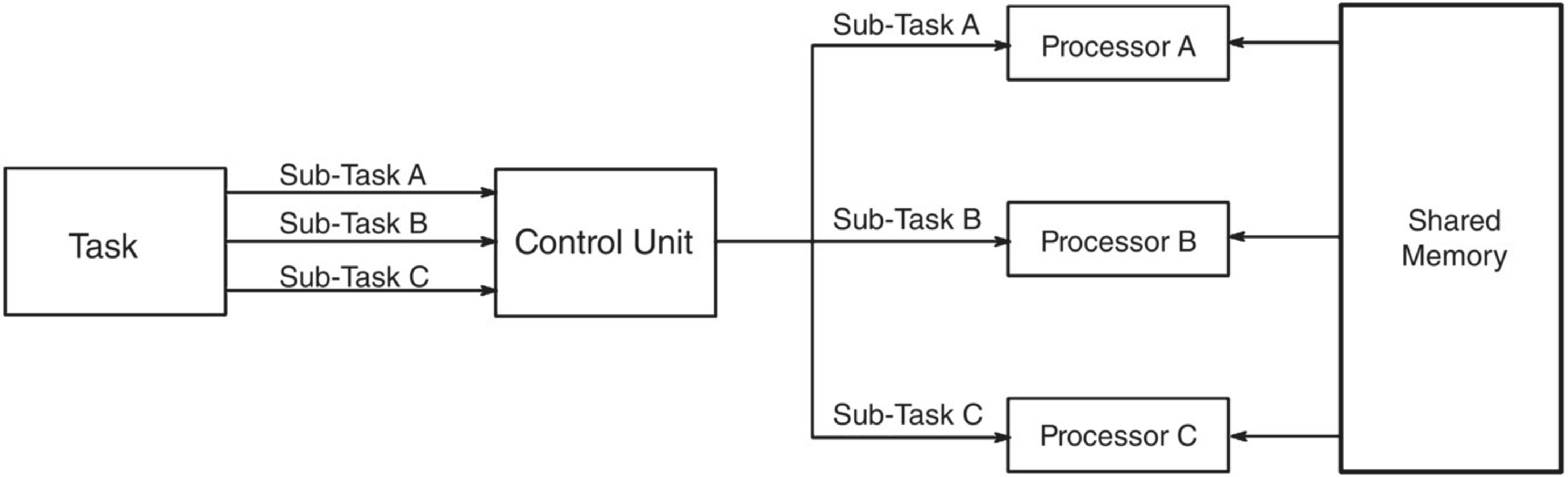
- Processes data as it arrives
- Low latency, continuous streams
- Example: Apache Storm, Spark Streaming



Solution	Developer	Type	Description
Storm	Twitter	Streaming	Framework for stream processing
S4	Yahoo	Streaming	Distributed stream computing platform
Mill Wheel	Google	Streaming	Fault tolerant stream processing framework
Hadoop	Apache	Batch	First open source framework for implementation of Map Reduce
Disco	Nokia	Batch	MapReduce framework by Nokia

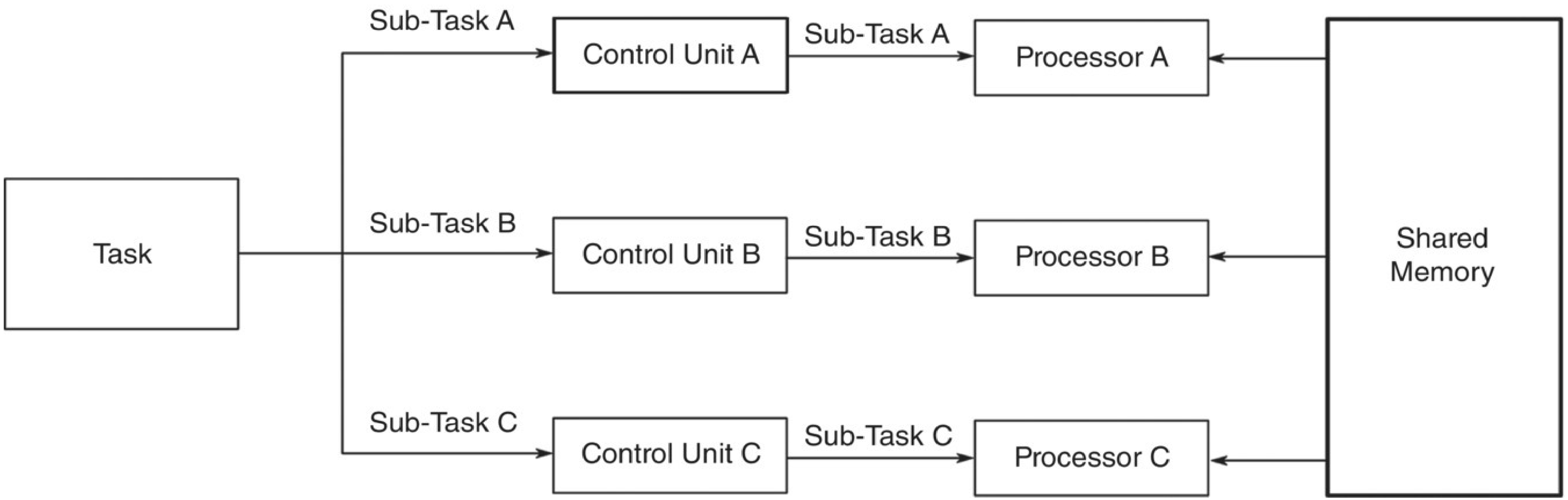
Parallel Processing

- Tasks are split across multiple processors
- Improves speed and efficiency
- Used in scientific and big data workloads



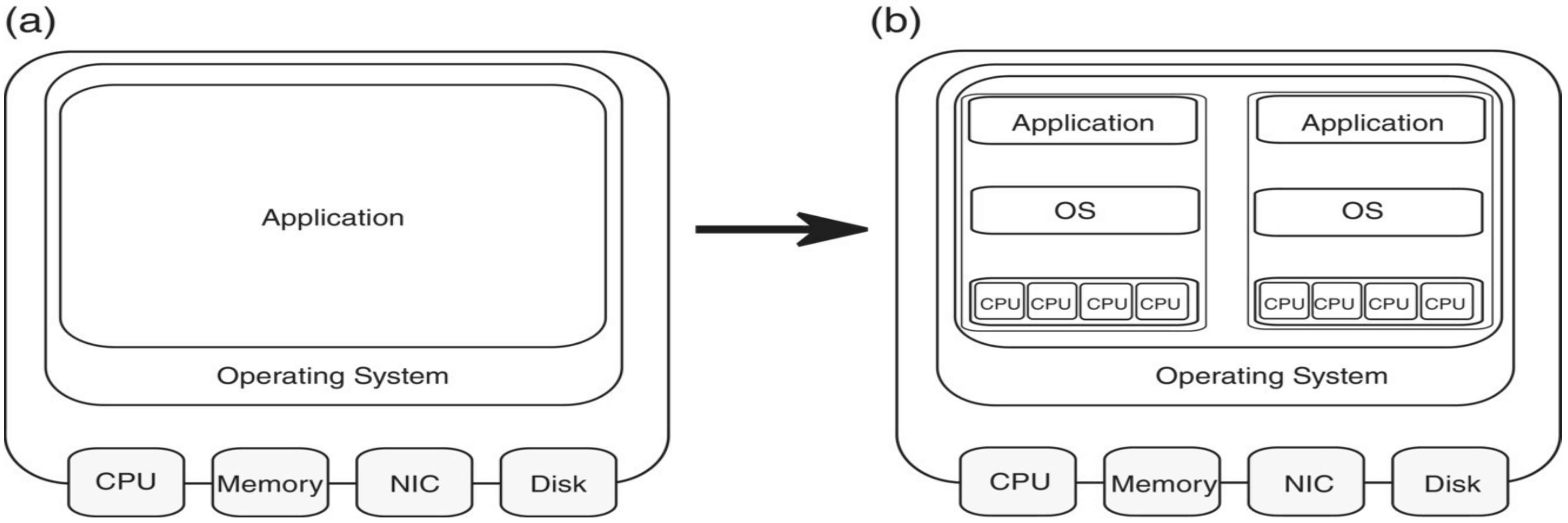
Distributed Computing

- Computation spread across multiple machines
- Improves scalability and reliability
- Foundation of modern big data systems



Big Data Virtualization

- Virtual machines share physical hardware
- Improves resource utilization
- Enables flexible deployment of big data platforms



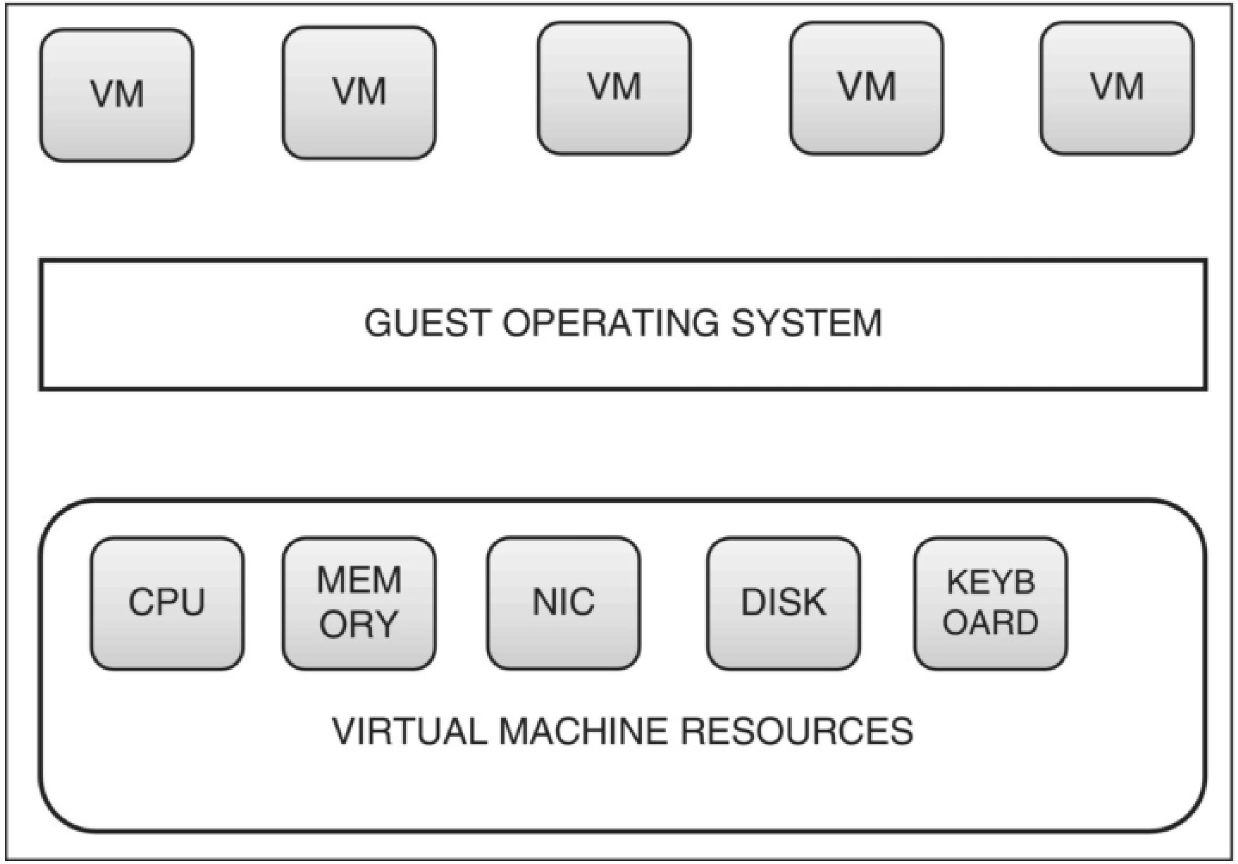
System architecture before and after virtualization.

Attributes of Virtualization

- Virtualization enables efficient use of physical hardware
- Core characteristics define how virtualization works
- **Attributes of Virtualization:**
  - **Encapsulation**:
    - A Virtual Machine (VM) is a software representation of a physical machine
    - VM is a software representation of a physical machine
  - **Partitioning:**
    - Physical hardware is divided into multiple logical partitions
    - Each partition runs a separate VM
  - **Isolation:**
    - Virtual machines are isolated from each other
    - VMs are isolated from the host physical system
    - Failure of one VM does not affect other

Big Data Server Virtualization

- Virtualization inserts a software layer over host hardware
- Multiple operating systems run simultaneously on one machine
- CPU and memory resources are virtualized
- Multiple applications run on a single server
- Supports large-scale and real-time Big Data analytics
- Handles unpredictable and high-volume data processing demands



Cloud Computing Concept

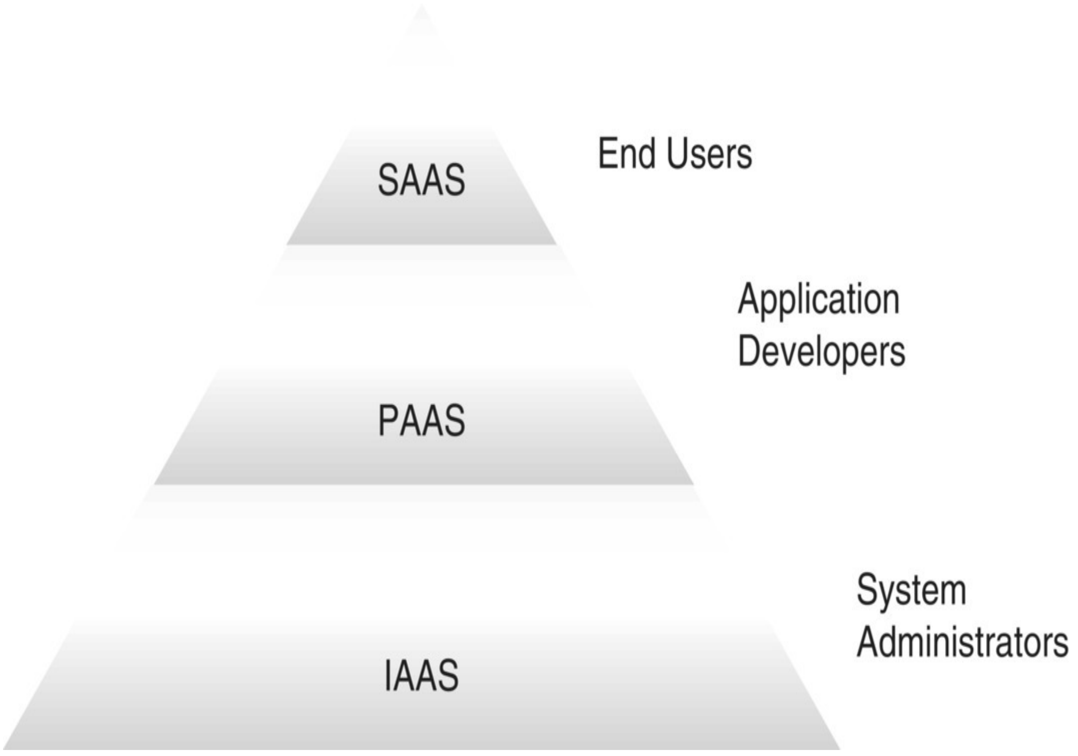
- **On-demand** access to computing resources
- Scalable and **pay-as-you-go** model
- Supports **big data storage and analytics**

Cloud Computing Types

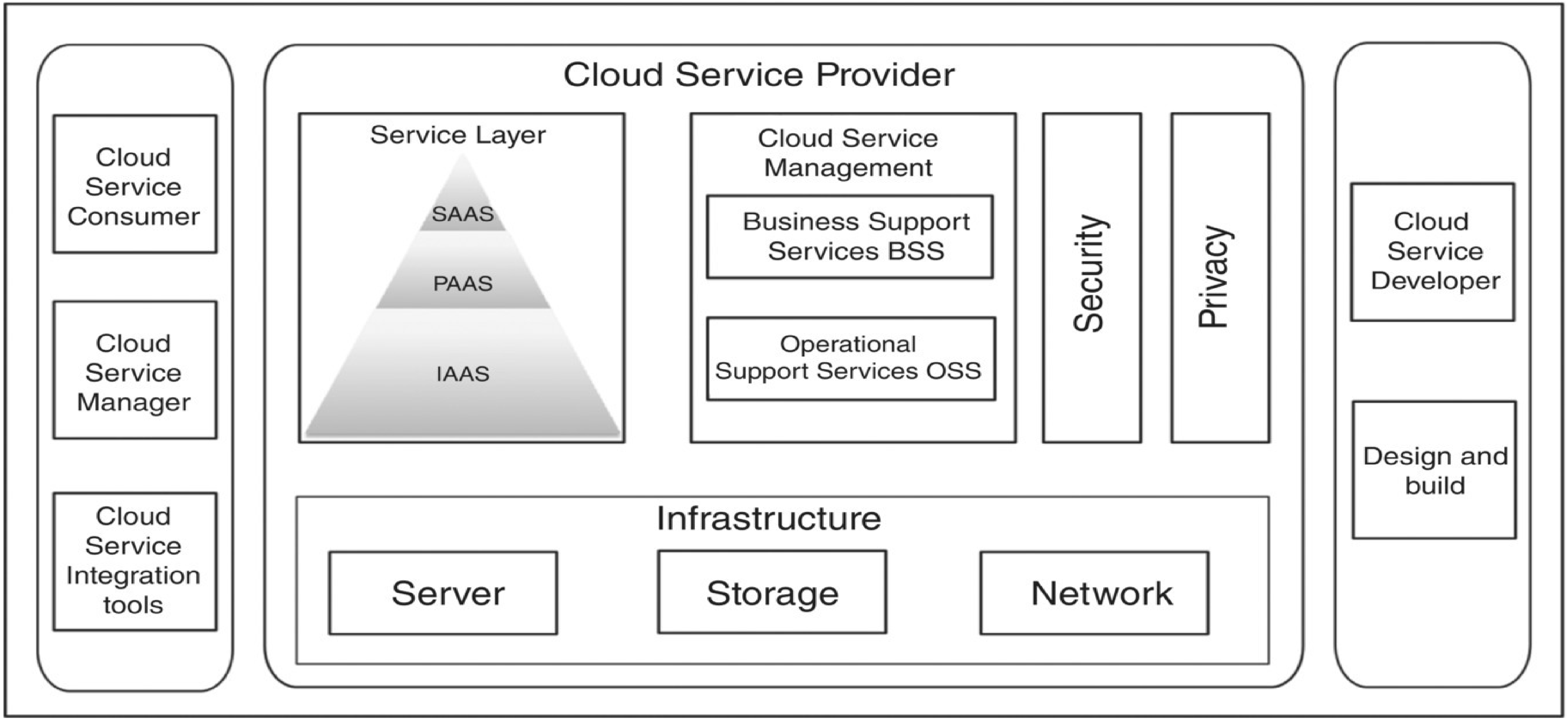
- **Public cloud**:
  - Services delivered over the Internet by third-party providers on a **pay-as-you-go** model
  - Offers **high scalability, low cost, and minimal maintenance**, but limited control and security
  - Example: iCloud, Google Drive, and playing music from Amazon’s cloud player are all public cloud services.
- **Private cloud**:
  - Cloud infrastructure **dedicated to a single organization**
  - Provides **greater security and control**, but involves **higher cost and maintenance**
- **Hybrid cloud:**
  - Combination of **public and private clouds**
  - Enables **flexibility, on-demand scalability**, and efficient handling of variable workloads

Cloud Services and Architecture

- **IaaS, PaaS, SaaS** service models
- Layered cloud architecture
- Supports elastic big data workloads
- **IaaS** provides access to the following resources on a pay-per-use basis:
  - Servers
  - Networking
  - Data center space
  - Load balancers
  - Storage

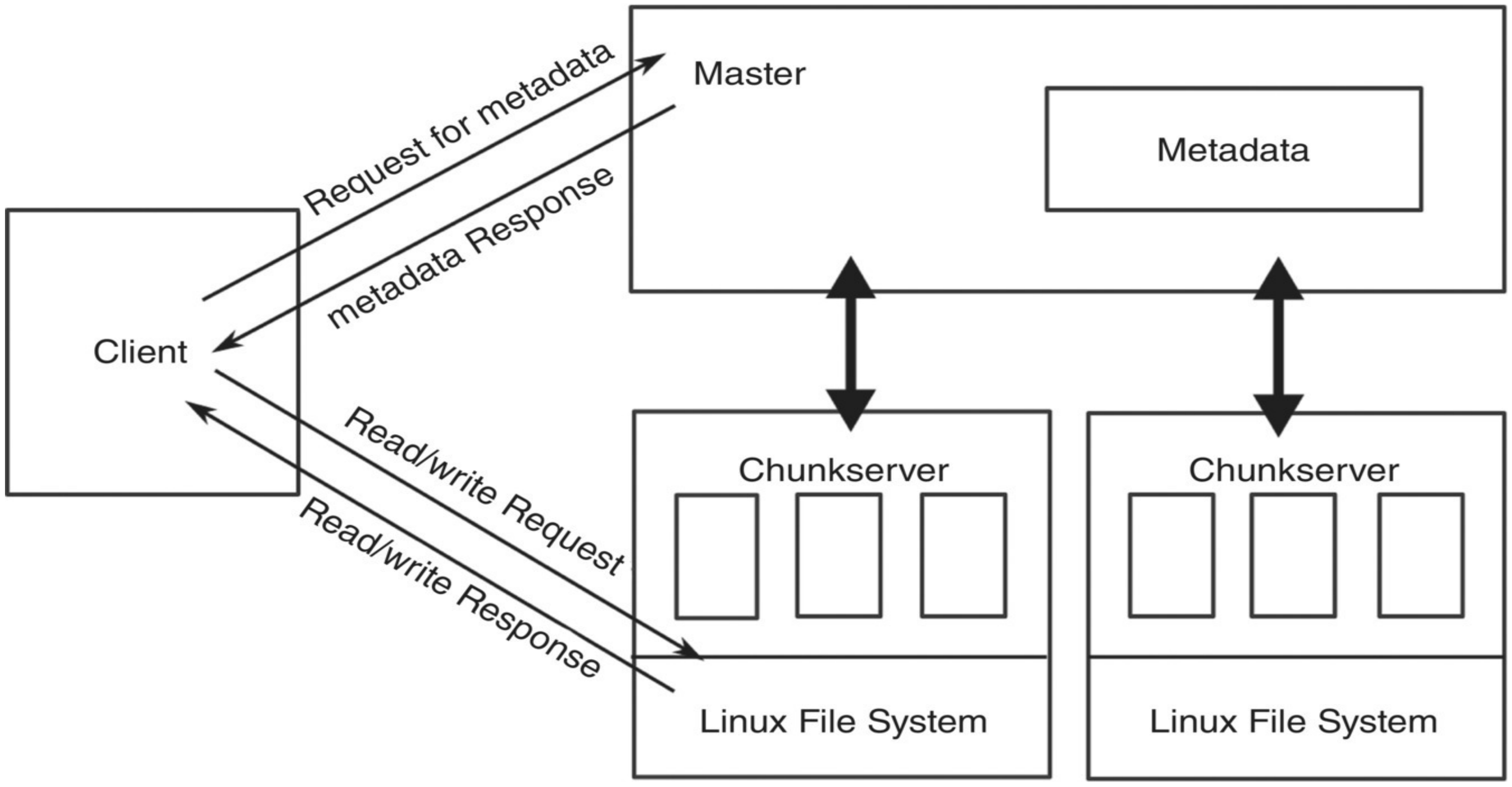


Cloud Architecture



Google File System

- Google File System (GFS) is designed for fault tolerance
- It is a foundation for modern cloud storage



Cloud Storages

- Big data storage requires s**calability, reliability, availability, cost-effectiveness, and fault tolerance**
- Cloud storage relies on **distributed file systems** for large-scale file storage and **NoSQL databases** for processing and analyzing massive data
- Cloud storage is a distributed storage for large-scale data
- Examples: Amazon S3 (AWS), Google Cloud Storage (GCS), Microsoft Azure Blob Storage, etc.

Cloud Challenges

- Security and Privacy
- Portability
- Computing performance
- Reliability and availability
- Interoperability

Reference/textbook

- Chapter 4-Part I & II of the TextBook: Big Data: Concepts, Technology, and Architecture, Balamurugan Balusamy, Nandhini Abirami R, Seifedine Kadry, Amir H. Gandomi, Wiley, 2021

Next Session: Big Data Tools & technologies

Q/A

Thank you!